a

Energy

Number frequency

1.0
0.8
0.6
0.4
0.2
0.0

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

Shallow, merged wells:
approximate readout

Deep, isolated wells:
exact readout

Evidence axis x

b

Entropy

$y_0=2$
$H=0.01$

1 2 3

$y_0=6$
$H=1.29$

5 6 7 8 9

$y_0=11$
$H=2.39$

9 11 13

Posterior distribution
over input y_0

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

c

Predicted RT (s)

2.5
2.0
1.5

Basal latency t_0
Attractor time $\alpha E[T]$
Readout uncertainty $\beta_H E[H]$

Slow, approximate

Fast, exact

$\mathrm{RT} = t_0 + \alpha E[T] + \beta_H E[H]$

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

Presented numerosity n